DATABASE USER MANUAL:

“ENERGY TRANSITION EVIDENCE IN CHILE 2025”

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1) Introduction

The “Energy Transition Evidence in Chile 2025” datasets project compiles key information on the main processes, challenges, and opportunities related to the country’s energy transition. This tool has been developed to serve as a platform for analysis and consultation, offering essential data to understand the evolution of the energy sector in Chile and to measure progress towards a cleaner and more efficient future.

In an effort to ensure the accuracy and traceability of the collected data, this database collects information from multiple institutional sources, such as the Ministry of Energy, the National Energy Commission, and the National Electrical Coordinator. The data is organized into categories that cover energy supply and consumption, greenhouse gas emissions, and energy efficiency indicators. The main objectives include:

- Consolidating information from multiple sources into a single repository allows the comparison of different factors related to energy transition and climate change mitigation strategies in Chile.
- Enabling the analysis of historical data and current trends in the country's energy sector, promoting evidence-based decision-making and the ability to compare data from different years to visualize the development of the energy transition in Chile.
- Providing an accessible tool for diverse users, including academics, decision-makers, students, and public and private institutions interested in the energy field.

The design and development of this database respond to the need for a robust repository that reflects the current state of Chile's energy sector and facilitates historical analysis of how this process has evolved.

2) Database Structure

The datasets compile open available information on energy projects in Chile and key energy transition indicators. To facilitate data analysis and ensure a coherent structure, the information has been organized following international references, such as the guidelines of the International Energy Agency (IEA)[1]. This approach ensures that the data is useful for analyzing phenomena related to the energy transition, both nationally and internationally.

The information in the datasets is grouped into a main master .xlsx file and related R files when size is too big for Excel to handle. The former is structured onto four main pillars, which allow for a comprehensive examination of the most relevant aspects of the energy transition in Chile:

- Progress of Renewable Energies in Chile.
- Energy Demand and Consumption.
- Energy Efficiency in Chile.
- Greenhouse Gas Emissions.

2.1) Main information sources

The database integrates information from various institutions selected for their relevance and reliability in the energy field. The main sources are presented in Table A1:

Table A1: Database information sources.

Source	Main Content	Time Period
National Energy Commission (CNE)	List of active energy projects,[2] Gross energy generation.	1996 - 2025
National Electric Coordinator (CEN)	Hourly generation by plant, historical installed capacity.	2000 - 2024
Open Energy	Primary and secondary energy matrices, energy consumption by sector, and energy intensity.	2008 - 2023
Ministry of the Environment	National GHG emissions inventory, GHG emissions inventory by region, particular breakdown of CO_2 and CH_4 emissions.	1990 - 2022

[1] International Energy Agency (IEA). (2023). *World Energy Outlook 2023: Executive Summary.*
[2] Note: The data for the “List of Active Projects” variable, covering the years 2017 to 2023, were collected through internal communication via the transparency portal in their website.

3) Datasets Construction Methodology

The construction of the datasets was carried out using a systematic approach that combined the extraction of original data with the necessary processing and normalization to ensure the consistency and usefulness of the information. In general terms, the tables included in the database are divided into two main categories: those retrieved directly from their original sources and those constructed through data integration and processing

3.1) Tables retrieved from original sources

These tables have been incorporated into the database without modifications to their content, preserving fidelity to the original records. This ensures traceability of the data and alignment with official sources. The following tables are part of this group:

Table A2: Detail of tables retrieved from original sources

Title	Original Source	Applications
Installed energy capacity	National Electrical Coordinator (CEN)	Evaluate the historical evolution of installed capacity in the country and analyze the contribution of different energy types to the country's energy mix.
Energy intensity	Open Energy	Measure the relationship between energy consumption and economic growth, evaluating national-level energy efficiency.
GHG emissions inventory	Open Energy	Monitor national greenhouse gas emissions, particularly from the energy sector.
GHG emissions inventory by region	Ministry of the Environment	Analyze the regional distribution of greenhouse gas emissions.
CO_2 Emissions	Ministry of the Environment	Evaluate the contribution of the energy sector to CO_2 emissions and its environmental impact.
CH_4 Emissions	Ministry of the Environment	Evaluate the contribution of the energy sector to CH_4 emissions and its environmental impact.
Energetic balance	National Energy Comission	Evaluate the historical evolution of energy supply and demand

3.2) Self-Elaborated Tables

These tables have been elaborated through the reorganization, integration, and consolidation of data from various sources. The processing of this data allows for a clearer and more adapted presentation of the database objectives. The processed tables are detailed below:

Table A3: Detail of self-elaborated tables

Title	Original source	Construction
List of Active Projects (2017 - 2025)	National Energy Commission (CNE)	**Initial Review**: Download and analysis of the complete active projects database for December of each year. **Variable Cleaning:** Elimination of redundant or irrelevant columns such as “Unit Code”, “Unit Type”, “Brand/Model”, etc. (23 in total) **Justification**: These variables were eliminated for being redundant with others present, not providing relevant information for the objectives of this database, or presenting high inconsistency. **Modification of the “Classification” variable”:** The categories 'NCRE' (Non-Conventional Renewable Energies) and 'Conventionals' were changed to 'Renewables' and 'Non-Renewables,' respectively.". **Justification:** To align terminology with database standards, using the terms 'Renewables' and 'Non-Renewables' as detailed in Table A5.
Gross Energy Generation (1996 - 2025)	National Energy Commission (CNE)	**Data Integration:** Consolidation of data from main electrical systems and minor subsystems in one table. **Note:** From 1996 to 2017 the data was disaggregated by SIC and SING systems. Since 2018, these systems were integrated with the creation of the SEN (National Electrical System). Additionally: ● Data from the Magallanes subsystem is available starting in 1998. ● Data from the Los Lagos subsystem is available starting in 2005. ● Data from the Aysén subsystem is available starting in 2008.

Title	Original source	Construction
Hourly generation by plant (2000 - 2024)	National Energy Commission (CEN)	**Data Integration:** Consolidation of data from main electrical systems and minor subsystems in one table. **Note:** Between 2000 and 2015, the available information corresponds only to the SIC subsystem, as the SING and minor subsystems did not present data broken down by the 24 hours of the day.
Primary energy mix (2008 - 2023)	National Energy Balance (Open Energy)	**Data Integration:** Consolidation of information from different years into a single table. **Construction of primary supply variables:** The total primary energy supply is calculated as the sum of national production and imports, minus exports and the stock variables of each resource.
Final energy consumption by sector (2008 - 2023)	National Energy Balance (Open Energy)	**Value Calculation:** Calculation of final consumption and breakdown by sector type (Transport, Industrial, Energy: Own Use, Residential, Mining, Commercial, and Public). **Classification:** Classification of resources according to the type of secondary energy resource presented in Table A 6.
Decarbonization index measures	National Energy Balance (Open Energy)	**Value Calculation:** Final energy consumption is converted to TJ and combined with IPCC default CO_2 emission factors to estimate annual carbon intensity. The Decarbonization Index (DCI) is then calculated as the reduction in carbon intensity relative to a hypothetical energy system supplied entirely by coal, following Qi et al. (2024). **Classification:** Individual fuels are harmonized into twelve DCI fuel categories. The index is further decomposed into DCINC, representing the contribution of non-carbon energy sources, and DCIOG, representing decarbonization achieved through substitution from coal towards lower-carbon fossil fuels.

4) Standardization and Classification of Variables

To ensure consistency in the analysis and presentation of data, standardization, and classification criteria have been established for the most relevant variables in the database. This section describes the methodologies used and provides key reference tables.

4.1) Energy Type Classification

During the construction of the database, the information was compiled from multiple sources, each with its categorization of "Energy Type." These differences generated inconsistencies when integrating them. Therefore, Table A4 presents a tool to standardize these categories and ensure consistency in the analyses.

Table A4: Adjustment factors for the “Energy Type” variable:

Final Classification	Original Classification
Coal	Coal, Coal- Petcoke, Petcoke.
Biomass	Biogas, Biomass, Biomass - Fuel Oil N°6
Natural Gas	Cogeneration, Natural Gas, Combined Cycle Thermal, Propane, Liquefied Petroleum Gas.
Wind Power	Wind Power
Diesel Oil	Fuel Oil No. 6, Diesel, Diesel Oil.
Small Run-of-River Hydro	Small Run-of-River Hydro
Dam Hydropower (Reservoir hydropower)	Dam Hydropower
Geothermal	Geothermal
Solar-CSP	Solar-CSP
Solar Photovoltaic	Solar Photovoltaic

4.2) Classification by Renewables and Non-Renewables

Within the database, the "Classification" variable is frequently used to divide energy sources into Renewables and Non-Renewables categories. This section provides Table A 5 as a reference to standardize both categories, ensuring consistency in their interpretation and analysis.

Table A5: Adjustment factors for the “Classification” variable.

Classification	Energy Types
Non-Renewable	Natural Gas, Diesel Oil, Coal
Renewable	Biomass, Wind Power, Solar Photovoltaic, Mini Run-of-River Hydro, Dam Hydro, Geothermal, Solar-CSP

4.3) Classification of Secondary Energy Resources

Secondary energy resources, defined as those derived from primary energy resource transformation processes, are fundamental to understanding the structure of energy consumption. Within the database, these resources have been classified to facilitate their analysis and standardization.

Table A 6 specifies each of the secondary energy resources measured in this classification, grouping them according to their origin and characteristics.

Table A6: Classification of Energy Resources in the Secondary Matrix

Category	Secondary Energy Resources
Oil products	Diesel oil, Fuel Oil, Motor Gasoline, Kerosene, Liquefied Gas, Aviation Gasoline, Aviation Kerosene, Naphtha, Refinery Gas, Petroleum middle distillates
Electricity	Electricity
Bioenergy	Biomass, Biogas, Black Liquor, Biomass Pellets
Natural Gas	Natural Gas

Coal and coke	Coal, Mineral Coke, Coke Oven Gas, Tar, BlastFurnace Gas, Petroleum Coke

5) Glossary of Terms

The glossary of terms included in this manual aims to provide clear and consistent definitions of the key concepts used in the database, facilitating user understanding and ensuring proper use of the information. In this case, two energy glossaries developed by the Ministry of Energy of Chile have been taken as a reference.

Table A7: Energy Glossary A[3]

Term	Description
Primary Energy	Energy sources in their natural state, meaning they have not undergone any type of physical or chemical transformation through human intervention. They can be obtained directly from nature (e.g., hydraulic energy, solar, wood, and other vegetable fuels) or after extraction processes like petroleum, coal, geo-energy, etc.
Secondary Energy	Energy products obtained through the transformation of primary sources or other secondary sources. These sources and forms of secondary energy are classified according to the primary source they come from. They include electricity, secondary petroleum products, coal derivatives, biomass derivatives, and biofuels.
Energy efficiency	Achieving the same result while consuming less energy, without diminishing the quality of life, or the quality of the products or services delivered.
Electrification	Process of substituting the use of fossil fuels with electricity as an energy input for the production of goods and services, for transport, or daily use.
Energy intensity	An indicator of the energy efficiency of an economy or process, calculated as the ratio of energy consumption to an activity indicator (such as GDP or value added). A common example at the country level is the energy consumed per unit of gross domestic product (GDP) generated, which allows for comparing energy performance between countries or over time.
Teracalorie	A unit of energy measurement equivalent to one trillion (10^{12}) calories, used to express large-scale energy consumption or production figures. For reference, the total primary energy supply of a country like Chile is measured in tens of thousands of teracalories annually. This unit makes it easier to manage and compare the vast amounts of energy involved in sectors like transport or industry.

[3] Ministry of Energy. 2021. *Transición Energética de Chile*. energia.gob.cl/sites/default/files/documentos/actualizacion_anteproyecto_pen_2021_lr.pdf

Table A8: Energy Glossary B[4]

Energy Resource	Description
Tar	A viscous, black, strong-smelling liquid obtained as a byproduct of the destructive distillation of coal (known as coal tar). Although sometimes confused with asphalt (or bitumen), which is a residue from petroleum refining, tar is chemically distinct. Historically, it has been used for waterproofing and as a binder in road construction.
Biogas	The gas, mainly methane, obtained from the anaerobic fermentation of biomass waste and sanitary landfills. It is used as fuel for internal combustion engines coupled with electricity generators.
Coal	A solid combustible mineral, black or dark brown, containing essentially carbon, as well as small amounts of hydrogen, oxygen, nitrogen, sulfur, and other elements. It results from the degradation of plant organism remains over long periods, through the action of heat, pressure, and other natural physical-chemical phenomena. Because there are different degrees of change in the process, mineral coal is not a uniform mineral and is classified by rank according to its degree of degradation, in series ranging from lignite to anthracite, through sub-bituminous and bituminous, which present considerable differences in their volatile matter content, fixed carbon, and calorific value. In terms of end-use, mineral coal can be divided into two classes: a) coking or metallurgical coal and b) steam or thermal coal.
Mineral Coke	Mineral coke (or metallurgical coke) is an energy byproduct obtained from a mineral coal distillation process. It provides heat for reduction reactions and the fusion of reduced material and slag.
Petroleum Coke (Petcoke)	A solid, porous, non-fusible fuel, generally black, with a high carbon content (90% - 95%), obtained as a residue from petroleum refining. It is used as an input in coking plants for the steel industry, in the manufacturing of electrodes, in chemical production, and as fuel for heating.
Diesel Oil	A liquid fuel obtained from the atmospheric distillation of petroleum between 200 and 380 degrees Celsius. It is heavier than kerosene and is used in diesel-type internal combustion engines (automobiles, trucks, power generation, marine and railway engines), as well as for heating in industrial and commercial applications.
Electricity	The energy transmitted by moving electrons. It includes electricity generated from any resource, whether primary or secondary, renewable or non-renewable, in different types of power generation plants.
Blast Furnace Gas	It is obtained as a byproduct of steel production in blast furnaces and is generally used as fuel for heating purposes within the plant.
Coke Oven Gas	The gas obtained as a secondary product from the intense heating of mineral coal or coke, with a mixture of air and steam, in coking plants. It is composed of carbon monoxide, nitrogen, and small amounts of hydrogen and carbon dioxide.
Town Gas	It corresponds to a mixture of biogas or propane enriched with LPG or natural gas. Its distribution medium is through the piped network from gas holders. It is used for cooking, water heating, and home heating, among other uses. A large part of this fuel's use is in the CPR (Commercial, Public, and Residential) sector.
Refinery Gas	Non-condensable gas, obtained from crude oil refining and composed mainly of hydrogen, methane, and ethane. It is used as an energy source in the refining process itself.

[4] Ministry of Energy. 2022. *Informe Balance Nacional de Energía 2020*. https://energia.gob.cl/pelp/balance-nacional-de-energia

Energy Resource	Description
Natural Gas	A gaseous mixture of hydrocarbons. It includes both free and associated natural gas and is also present in coal mines or geopressured zones. For energy balance purposes, they are considered as a single source, both the free gas and the net-produced associated gas, due to their similar nature and uses.
Motor Gasoline	A mixture of light, liquid hydrocarbons obtained from petroleum distillation and/or natural gas treatment, with a boiling range generally between 30-200 degrees Celsius. This group includes 93, 95, and 97 octane gasolines.
Aviation Gasoline	A mixture of reformed naphthas with a high octane rating, high volatility, stability, and a low freezing point, used in propeller aircraft with piston engines.
Liquefied Petroleum Gas (LPG)	Consists of a mixture of light hydrocarbons obtained as products of refining processes, crude oil stabilization, and the fractionation of natural gas liquids. It can be of three types: - Mixture of C3 group hydrocarbons (propane, propene, propylene) - Mixture of C4 group hydrocarbons (butane, butene, butylene) - Mixture of C3 and C4 in any proportion LPG is normally liquefied under pressure for transport and storage.
Kerosene	A fuel used for cooking, lighting, engines, refrigeration equipment, and as a solvent for asphalt and domestic insecticides.
Aviation Kerosene	A liquid fuel composed of the petroleum fraction that distills between 150 and 300 degrees Celsius. It has a special degree of refining that gives it a lower freezing point than common kerosene. It is used as fuel in jet engines and turboprops.
Methanol	A petrochemical liquid composed of four parts hydrogen, one part oxygen, and one part carbon. It is a substance that occurs naturally and can be produced from both renewable and non-renewable fossil fuels (e.g., natural gas) that contain carbon and hydrogen.
Naphta	A light petroleum fraction obtained by direct distillation between 30 and 210°C. It is used as a feedstock for both gasoline manufacturing and the petrochemical industry. It is also used as a solvent in the chemical industry (manufacturing of paints and varnishes).
Fuel Oil	Residual fuel from petroleum refining that comprises all heavy products (including those obtained by blending). It is generally used in boilers, power generation plants, and engines used in navigation.
Crude Oil	A complex mixture of hydrocarbons of varying molecular weight, in which there is a generally small fraction of compounds containing sulfur and nitrogen. The composition of petroleum is variable and can be divided into three classes according to the distillation residues: paraffin, asphalts, or a mixture of both. In its natural state, it is in a liquid phase and remains liquid under normal pressure and temperature conditions, although in the reservoir, it may be associated with gaseous hydrocarbons. This concept includes liquids from associated gas that condense upon reaching the surface, in production facilities (petroleum condensates), or other liquid hydrocarbons that are mixed into the commercial crude oil stream. Crude oil is the main feedstock for refineries for the production of petroleum products or derivatives.

6) Codebook

Table A.1: List of Variables

Title	Variables	Unit of Measurement
List of Active Projects	• project_name • owner • classification • energy_type • region • municipality • system • comissioning_year • interconnection_point • easting_utm • northing_utm • utm_zone • gross_power_mw • net_power_mw	Gross Power: Megawatt [MW] Net Power: Megawatt [MW]
Gross Generation	• year • biomass • coal • coal_petcoke • cogeneration • wind_power • fuel_oil • natural_gas • lng • dam_hydropower • run_of_river_hydropower • small_run_of_river_hydropower • petcoke • diesel_oil • diesel_oil_and_fuel_oil • geothermal • solar_photovoltaic • concentrated_solar_power • total	Gigawatt-hour [GWh]
Installed Capacity	• year • hydro • coal • diesel_oil • natural_gas • wind_power • solar_photovoltaic • concentrated_solar_power • geothermal • other • total	Megawatt [MW]
Hourly Generation by Power Plant	• power_plant_name • energy_type • energy_subtype • region • date • h_(1:24)	Megawatt-hour [MWh]
Primary Energy Matrix	• año • combustible • fuel • tcal	Teracalorie [Tcal]

	○	
Final Energy Consumption by Sector (Detailed)	● year ● sector_consumo ● fuel ● tcal	Teracalorie [Tcal]
Energy Intensity	● nominal_gdp_2018 ● energy_supply ● energy_intensity	Nominal GDP 2018: [Billions of CLP] Energy Supply: Teracalories [Tcal] Energy Intensity: [Tcal/Billion CLP]
GHG Emissions Inventory	● year ● energy_industries ● manufacturing_industries_and_construction ● transport ● other_sectors ● unspecified ● fugitive_emissions_from_fuels ● co2_transport_and_storage ● industrial_processes_and_product_use ● agriculture ● waste ● total_inventory	Megatonnes of CO_2 equivalent [$MtCO_2Eq$]
Regional GHG Emissions Inventory	● year ● arica_parinacota_region ● tarapaca_region ● antofagasta_region ● atacama_region ● coquimbo_region ● valparaiso_region ● metropolitana_region ● ohiggins_region ● maule_region ● nuble_region ● biobio_region ● araucania_region ● los_rios_region ● los_lagos_region ● aysen_region ● magallanes_region ● total_inventory	Megatonnes of CO_2 equivalent [$MtCO_2Eq$]
CO_2 and CH_4 Emissions and Absorptions	● year ● energy ● industrial_processes_and_product_use ● agriculture ● land_use_change_and_forestry ● waste ● total_national_emissions ● total_national_emissions_and_removals	Kilotonnes [kt]
Balance	● año ● item ● categoria ● seccion	Teracalorie [Tcal]

	• combustible • tcal	
Decarbonization Index	• año • total_energy_tj • total_emisions • CI_t • DCI • DCINC • DCIOG	Terajoules (TJ)
DCI factors	• fuel_dci • fe_CO2_tCO2_TJ	

Table A.2: R cleaning and analysis variable's codebook

Variable	Type	Units	Description
año	Integer	Year	Calendar year (2008–2024)
year	Integer	Year	Calendar year
item	Character		BNE accounting block
categoria	Character		Original BNE category
seccion	Character		Original BNE section
combustible	Character		Original fuel
fuel	Character		Harmonized fuel
sector_consumo	Character		Harmonized consumption sector
tcal	Numeric	Tcal	Energy
tj	Numeric	TJ	Energy converted from Tcal
fuel_dci	Character		DCI fuel class
fe_CO2_tCO2_TJ	Numeric	tCO2/TJ	IPCC emission factor
total_energy_tj	Numeric	TJ	Annual energy
total_emissions	Numeric	tCO2	Annual emissions
CI_t	Numeric	tCO2/TJ	Carbon intensity
DCI	Numeric	Index	Decarbonization Index
DCINC	Numeric	Index	Non-carbon contribution
DCIOG	Numeric	Index	Oil and gas contribution
indicator	Character		Indicator name
value	Numeric	Index	Indicator value
energy_share	Numeric	Fraction	Energy share
emissions_share	Numeric	Fraction	Emission share
power_plant_name	Character		Plant name
energy_type	Character		Technology
energy_subtype	Character		Technology subtype
region	Character		Administrative region
date	Date	Date	Observation date
month	Integer	Month	Calendar month
month_name	Character		Month name
day	Integer	Day	Day of month
weekday	Ordered factor		Weekday

hour	Integer	Hour	Hour of day
hour_num	Integer	Hour	Numeric hour
mw	Numeric	MW	Hourly generation
mean_mw	Numeric	MW	Mean hourly generation
gwh	Numeric	GWh	Electricity generation
mean_gwh	Numeric	GWh	Mean hourly generation
share	Numeric	Fraction	Share